

Solutions to Feb (25)

a) If $\text{char } F = 0$ then $\mathbb{Z} \hookrightarrow F$ so F^* is not periodic
 so we may assume that $\text{char } F = p > 0$ and hence

$\mathbb{Z}/p\mathbb{Z} = F_p \subset F$. Let $x \in F^*$ then as F is periodic, $x^n = 1$
 some $n > 0$, so $[F_p[x] : F_p] \leq n$, so $F_p[x] \cong F_{p^r}$ some $1 \leq r \leq n$.
 So $x \in F_p^a \subset F^a$ for all $x \in F$ so $F \subset F_p^a$.

2) $x^3 - 3x^2 - 3$ i) $F = \mathbb{C} \Rightarrow G = \langle 1 \rangle$ is the trivial group & $|K| = 1$

ii) $F = \mathbb{R} \rightarrow \begin{array}{c} \swarrow \\ -5 \end{array} \Rightarrow x^3 - 3x^2 - 3 \text{ has 1 real root} \Rightarrow |K| = 2 \Rightarrow G = \mathbb{Z}/2\mathbb{Z}$

iii) $F = \mathbb{Q}$ $f(x)$ is irreducible by Eisenstein. $p(x+1) = x^3 - 3x - 5$

$D = -4(-3)^3 - 27(-5)^2 = 9^2(-7)$ not a square in $\mathbb{Q} \Rightarrow G = S_3$

iv) $F = \mathbb{Z}/2\mathbb{Z}$ $f(x) = x^3 + x^2 + 1$ has no roots so it's irreducible

So α a root of $f(x) \Rightarrow F_2(\alpha) \cong F_8 \Rightarrow |K| = F_8$ & $G = \mathbb{Z}/3\mathbb{Z}$

c) It suffices to find 2 irreducible polynomials of degree 3 with Δ not a square in $\mathbb{Q}$ and disjoint splitting field
 For example $(x^3 - 3x + 1)(x^3 - x - 1/3)$. It seems hard to
 check that $K_1 \cap K_2 = \mathbb{Q}$ i.e. that $[K_1 K_2 : \mathbb{Q}] = 9$.

Alternatively, consider $n = 91$, $\zeta = \zeta_n$ a primitive 77 root of 1.

Then $[\mathbb{Q}(\zeta_n) : \mathbb{Q}] \cong (\mathbb{Z}/91\mathbb{Z})^* \cong (\mathbb{Z}/7\mathbb{Z})^* \times (\mathbb{Z}/13\mathbb{Z})^* \cong \mathbb{Z}/6\mathbb{Z} \times \mathbb{Z}/12\mathbb{Z}$

is abelian, so $\mathbb{Q}(\zeta_n)^H = K$ is an abelian Galois ext of $\mathbb{Q}$ with
 group $(\mathbb{Z}/6\mathbb{Z} \times \mathbb{Z}/12\mathbb{Z})/H = G$. Pick $H = 2\mathbb{Z}/6\mathbb{Z} \times 4\mathbb{Z}/12\mathbb{Z}$ then

$G \cong \mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$.

d) We know that $[Q(\zeta_m):Q] = \varphi(m)$ & $\varphi(p^r) = p^r - p^{r-1}$

~~E contains infinitely many ζ_m , then~~

The set $\{\varphi(m) \mid \zeta_m \in E\}$ is also bounded by $[E:Q]$

Let $n = p_1^{r_1} \cdot \dots \cdot p_m^{r_m}$ then $\varphi(n) = (p_1^{r_1} - p_1^{r_1-1}) \cdot \dots \cdot (p_m^{r_m} - p_m^{r_m-1}) \in [E:Q]$

and it is easy to see that n is bounded so

there are only finitely many ζ_m in E .

e) $f(x) = x^3 - x + 1$ $f'(x) = 3x^2 - 1$

$$23 = (x^3 - x + 1)(18x + 27) + (3x^2 - 1)(-6x^2 - 9x + 4)$$

So if $\text{char } K \neq 23$, there are no repeated roots.

If $\text{char } K = 23$ then $(x^3 - x + 1) = (x - 13)^2(x + 3)$

So $f(x)$ always has 13 as a double root.